Sina Parsnejad

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Work Authorization: U.S. Permanent Resident (**Green Card holder**)

Social: www.linkedin.com/in/sina-parsnejad | Portfolio: https://sparsnejad.github.io/Portfolio.pdf

PERSONAL STATEMENT

Ph.D.-trained **Analog IC Design Engineer** with expertise in **mixed signal circuits, power electronics, sensors, and brain-computer interfaces.** Industry experience at Texas Instruments includes the full IC lifecycle from specification and design to simulation, validation, and post-silicon testing on proprietary CMOS technologies. Proficient in **Cadence Design Suite, MATLAB, Verilog, and embedded systems,** with a strong record of translating advanced research into reliable, high-performance semiconductor solutions. Passionate about developing **next-generation solutions** that bridge the physical and digital worlds.

EXPERIENCE

Freelance Artificial Intelligence Researcher, *Mercor,* 2025–Present

- Trained a generative AI model to tackle Olympiad-level engineering questions, enhancing problem-solving capabilities.
- Developed complex engineering questions, ensuring they met rigorous academic standards for evaluation.
- Graded and corrected answers to provide accurate feedback, fostering a deeper understanding of engineering concepts.

Analog IC Design Engineer, *Texas Instruments,* 2022–2025

- Designed and developed isolated and non-isolated half-bridge driver circuit IPs on TI's proprietary 90nm technology
- Successful chip Tape-outs (20 total): UCC571xx, UCC276xx, UCC274xx, UCC278xx
- Performed pre-silicon statistical and top-level sims, coordinated circuit layout implementation and post-layout sims, conducted automotive aging compliance testing, and in-laboratory post-silicon verification for low-side driver ICs
- Conducted quality assurance on TI projects, identifying and resolving design issues by performing post-silicon failure analysis and providing customer support for returned products
- Authored detailed design documentation, simulation reports, and design reviews for project milestones
- Performed outreach program as TI liaison for IEEE conferences and colleges

Research Assistant at *HAT lab,* *Michigan State University,* 2014–2022

- Project manager for development of peripheral nerve stimulation device for human augmentation
- Performed human trials to on 40 people, exploring cognitive effects of human augmentation techniques
- Developed portable electrochemical instruments using EDA PCB tools and performed field tests
- Developed a high-precision interface for arrays of nano-pore for single-molecule sensing, using ON 0.6µm
- Coordinated cross-functional collaboration among chemical science, software, electrical, and mechanical engineering teams to support the development of NSF and NIH grant proposals

Venture Fellow, *Spartan Innovation Center,* 2018–2019

- Market research and business plan development for MSU startup company, MagPlasma, Inc
- Participated in pitch competitions and investor meetings to present and promote potential business solutions

Research Assistant at *Beta* Laboratory, *Boğaziçi University,* 2012–2014

- Researched, designed, and fabricated three distinct chips in UMC 180µm technology, using Mentor IC design suite
- Designed a continuous-time Sigma-Delta ADC with a 25kHz bandwidth, achieving power consumption below 5µW
- Led a team of graduate students and coordinated individual VLSI design groups in UMC 0.5µm

EDUCATION

Michigan State University, Ph.D. of Electrical Engineering, East Lansing, Michigan

• Supervisor: Professor Andrew J. Mason • July 2017–August 2022
• Thesis subject: Human augmentation using unobtrusive rapid real-time machine-to-human communication

Michigan State University, M.S. of Electrical Engineering, East Lansing, Michigan

• Supervisor: Professor Andrew J. Mason • September 2014–July 2017
• Thesis subject: Power-area-efficient rapid-response CMOS frontend for high-throughput ion channel sensor array microsystems

SKILLS

IC design, using Cadence Design Suite and Mentor Graphics design suites from conception stages to statistical design, test, post-silicon in-lab validation. Coding in MATLAB, C, Python, Verilog. Proficient in ARM embedded system design, PCB design, and Jira change management. Communicative and proficient in English, Persian (Farsi), Turkish, Azerbaijani; intermediate Japanese

ADDITIONAL EXPERIENCE

- Associate editor for IEEE Open Journal of Circuits and Systems (OJCAS), 2026-2027
- IEEE MWSCAS 2025 conference Chair of Tutorials
- Invited presentation at IEEE Colombian BioCas conference, 2020 and NCSU University ASSIST / ICons Speaker series, 2025

PUBLICATIONS

- "A review of electrotactile stimulation for machine-to-human communication" IEEE TBME, 2026
- "Distinguishable Multi-layer Frequency-modulated Waveforms for Peripheral Nerve Communication in Human Augmentation Applications" IEEE Haptics, 2026
- "Etacteme Machine-to-Human Communication A Linguistically-Inspired Experimentally-Derived Electrotactile Icon Generation Scheme" Nature npj biomedical innovations, 2026 (In review)